

Reading-H class-wide selection as a conditional theorem: the assembled T6 promotion

TECT verification-first repository · claim B1-RH-ENUM

first issued 2026-06-06 · this version issued 2026-06-06 · v2.3

1. Purpose and scope

This note assembles the class-wide Reading-H selection as a conditional theorem and records the operator-authorized promotion of B1 from T5 (estimator-grade enumerated closure) to T6 CONDITIONAL on the named hypothesis set {H-LAYER, H-ADM-COH, SC-SCOPE} (H-A0 has been demoted to a verified anchor dependency: $m^* > m_w$, $M_R > M_c$ under A1-KERNEL-CONV at $\mu^2 = 0.005$). The statement is the selection SIGN: among all admissible competitors of the H-ADM-COH amended class, the isotropic Reading-H reference has the lowest free energy at matched second-cumulant scope. The quantitative selection margins remain estimator-grade (ESTIMATOR-UPGRADE, open) and are NOT part of the T6 claim.

2. The assembled theorem

Theorem 1 (Reading-H class-wide selection, conditional). *Assume H-LAYER (the comparison is the isotropic Gaussian-Hartree variational layer; within the diagonal-Gaussian class the isotropic dressing is the infimum, Prop A / Math427; the $A = 0$ uniqueness / zero-at-gap, former H-A0, is now a sign-decomposition THEOREM resting on the verified anchor dependency $m^* > m_w$, $M_R > M_c$ at $\mu^2 = 0.005$), H-ADM-COH (the coherence-resolution admissible-competitor class, operator verdict #14), and SC-SCOPE (matched second-cumulant order). Then for every admissible competitor pattern P ,*

$$F_{\text{total}}[P] \geq F_{\text{total}}[R_H],$$

with equality only for the Reading-H representative (the zero-equivalent / sub-resolution-equivalent configuration); on the quotient by sub-resolution equivalence the inequality is strict. Reading-H is selected. The proof is a case split on the competitor amplitude.

Proof (case split). Write $F_{\text{total}}[P] = F_{\text{layer}}[P] + \delta F_{\text{off}}[P]$, the diagonal-Gaussian layer energy plus the beyond-layer off-diagonal correction.

Bulk regime. By Prop A (H-LAYER; $A = 0$ uniqueness now a theorem), the isotropic layer is the infimum of the diagonal-Gaussian class with margin $F_{\text{layer}}[P] \geq F_{\text{layer}}[R_H] + m$, $m = +4.32 \times 10^{-3}$ the weakest interval band. By the STEP-5B closure on the H-ADM-COH amended class, $\delta F_{\text{off}}[P] > -m$ (margins $\times 59.4 / \times 8.8 / \times 2.6$ floors across the anchor band, AddA–AddE). Hence $F_{\text{total}}[P] \geq F_{\text{layer}}[R_H] + m + \delta F_{\text{off}}[P] > F_{\text{layer}}[R_H] = F_{\text{total}}[R_H]$.

Near-gap small-amplitude regime. Where the beyond-layer expansion is delicate (small-amplitude patterns near the gap, the G-U1-SMALLT corner), the structural floor applies directly:

at fixed total intensity I the diagonal Hartree dressing $\hat{r}(I) = r_R + 2\lambda'I$ is pattern-independent, so R_H and P share D_0 and

$$F_{\text{fluct}}[P] - F_{\text{fluct}}[R_H] = \frac{1}{2} \text{Tr} [\ln(D_0 + \lambda'P^2) - \ln D_0] \geq 0$$

unconditionally ($\lambda'P^2 \succeq 0$; $\ln$ operator-monotone; R-001, R-U10-3). The convention remainder that the verification triage mis-read as a $\times 2$ thinning is common-mode and cancels (neargap-common-mode-resolution, $\times 282$ margin dominance). The two regimes overlap and cover the amended class; the selection holds on both. $\square$

Lemma 1 (Equality classification). *In the near-gap case, $\frac{1}{2} \text{Tr} [\ln(D_0 + \lambda'P^2) - \ln D_0] = 0$ iff $\lambda'P^2 = 0$, i.e. $P = 0$ (the Reading-H representative). Indeed $\ln$ is strictly operator-monotone on positive operators, so for $\lambda'P^2 \succeq 0$ the trace difference is ≥ 0 with equality only when the perturbation vanishes on the support of $D_0^{-1/2}(\cdot)D_0^{-1/2}$; since $D_0 \succ 0$ this forces $\lambda'P^2 = 0$. Hence on the quotient by sub-resolution equivalence (which identifies $P = 0$ with the Reading-H representative) the selection inequality is STRICT for every nontrivial competitor. Appendix A states the trace-class hypotheses that make this rigorous in the renormalised per-volume setting.*

Case-split partition (explicit). Let $M(P)$ be the cell-averaged variance of the pattern's dressing. Define

$$\mathcal{A}_{\text{bulk}} := \{P \in \mathcal{A}_{\text{adm}} : M(P) \text{ in the Prop-A band/subband intervals where } m > 0\},$$

$$\mathcal{A}_{\text{near}} := \{P \in \mathcal{A}_{\text{adm}} : \text{small-amplitude near-gap section, } \|P\| \text{ below the band-floor crossover}\}.$$

By the Prop-A band structure $\mathcal{A}_{\text{adm}} = \mathcal{A}_{\text{bulk}} \cup \mathcal{A}_{\text{near}}$, and on the overlap BOTH the bulk inequality ($F_{\text{layer}} \geq F_{\text{layer}}[R_H] + m$, $\delta F_{\text{off}} > -m$) and the near-gap floor (≥ 0) hold, so the two cases do not double-count (each is a self-contained one-sided bound on the SAME normalised $F_{\text{total}}[P] - F_{\text{total}}[R_H]$). The crossover is formalised as a sign-invariant covering partition in Appendix B; its exact level remains estimator-grade but enters neither bound nor the sign verdict.

3. What the promotion does and does not assert

The T6 claim is the selection SIGN. It does NOT assert: (i) the precise quantitative margins (ESTIMATOR-UPGRADE, open); (ii) all-orders validity (SC-SCOPE is second-cumulant; the third-cumulant endpoint is marginal, U4/U15); (iii) unconditional closure (H-ADM-COH restricts the class; the unrestricted class is the DR-2 route, research-grade); (iv) μ^2 -neighbourhood robustness (ROBUSTNESS-MU2, open; pinned to $\mu^2 = 0.005$). Each is a named hypothesis or a tracked open gate.

4. Numerical verification

The promotion rests on previously certified artefacts (no new numerics): STEP-5B closure ([codes/vacuum/beyond_layer_gershgorin_bound.py](#) v1.14.0, 192/192) and the near-gap floor ([codes/vacuum/neargap_common_mode_repair.py](#) v1.0.0, 14/14).

Ingredient	Source	Status
layer margin $m = +4.32 \times 10^{-3}$	Prop A (B2, H-LAYER)	T6 on {H-LAYER}
beyond-layer $\delta F_{\text{off}} > -m$	STEP-5B closure (AddA-AddE)	CLOSED-CONDITIONAL (H-ADM-COH)
near-gap floor ≥ 0	R-001 / R-U10-3	unconditional (14/14)
scope	second cumulant	SC-SCOPE (substantive)

Quantitative sanity check (mandatory): the bulk inequality requires only $\delta F_{\text{off}} > -m$, and the weakest STEP-5B floor is $\times 2.6$ at $I = 2 \times 10^{-3}$, i.e. $|\delta F_{\text{off}}| < m/2.6 < m$ — strict with room; the near-gap floor is a separate non-negative term, so the two cases cannot both be tight simultaneously.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3.1 + 6.3.5(a))

- (α) “*The chain mixes two comparisons — the layer margin is each-state-at-its-own-dressing, the floor is common-dressing. Is the composition consistent?*” VALID-with-mitigation: the two are not summed; they are a CASE SPLIT over disjoint-but-covering amplitude regimes (bulk vs near-gap). Each case is a self-contained inequality with its own reference. The case split is stated explicitly in the proof; no double-counting occurs.
- (β) “*SC-SCOPE is not a formality: U_4/U_{15} found the third-cumulant endpoint corrections marginal ($\times 0.97/\times 1.0$).*” VALID — which is exactly why SC-SCOPE is a NAMED hypothesis, not a silent assumption. The T6 theorem holds at second-cumulant order; the all-orders lift (per-transfer third-cumulant kernels at the endpoint) is the registered open target. The promotion does not claim it.
- (γ) “*ESTIMATOR-UPGRADE is open — the quantitative ΔF figures are estimates, so how can this be T6?*” DISMISSED with the scope distinction: T6 here is the selection SIGN (a chain of inequalities: $m > 0$, $\delta F_{\text{off}} > -m$, $\text{floor} \geq 0$), each theorem-grade in its hypothesis. The precise margin VALUES are estimator-grade and tracked separately; they do not enter the sign.
- (δ) “*H-A0 was itself only B2’s T6 hypothesis — circular promotion?*” DISMISSED: the dependency DAG is acyclic ($B1 \rightarrow B2 \rightarrow A1$; no back-edge). B1’s T6 inherits B2’s named hypotheses {H-LAYER} forward (H-A0’s $A = 0$ uniqueness is now a theorem, demoted to a verified anchor dependency), which is composition, not circularity. The linter’s tier-monotonicity rule is satisfied: B1’s sub-T6 hard dependency A1 (T5) is covered by the named hypothesis set.
- (ϵ) “*Promoting the ‘enumerated’ claim B1 to a class-wide statement overwrites its precise scope.*” VALID-with- mitigation: the scope field is rewritten to state the class-wide conditional selection explicitly, and the enumerated estimator-grade figures are retained as the ESTIMATOR-UPGRADE open item. The claim statement (“Reading-H is selected”) is unchanged; only its tier and evidential basis advance.

No objection is UPHELD as a blocker; two are DISMISSED, three are VALID-with-mitigation (each a named hypothesis or tracked open item).

6. Appendix A: trace-log equality in the determinant class

Lemma 2 (Trace-log equality, determinant class). *Let $D_0 \succ 0$ be the common diagonal Hartree dressing and $A := D_0^{-1/2} \lambda' P^2 D_0^{-1/2}$. Assume (H1) $A \succeq 0$ and (H2) A is trace-class on the renormalised fluctuation space, i.e. $\sum_j a_j < \infty$ for the eigenvalues $\{a_j\} \subset [0, \infty)$ of A — the same finiteness that makes $\Delta F = \frac{1}{2} \text{Tr}[\ln(D_0 + \lambda' P^2) - \ln D_0] = \frac{1}{2} \text{Tr} \ln(\mathbf{1} + A)$ finite (certified for the admissible class in neargap-common-mode-resolution). Then*

$$\text{Tr} \ln(\mathbf{1} + A) = \sum_j \ln(1 + a_j) = 0 \implies a_j = 0 \ \forall j \implies A = 0,$$

and since $D_0^{-1/2}$ is invertible, $\lambda' P^2 = 0$, i.e. $P = 0$.

Proof. (H1): $\lambda' > 0$, $P^2 \geq 0$, and congruence by the self-adjoint invertible $D_0^{-1/2}$ preserves positivity, so each $a_j \geq 0$. Each summand $\ln(1 + a_j) \geq 0$ with equality iff $a_j = 0$. A convergent sum (H2) of non-negative terms vanishes iff every term vanishes; hence all $a_j = 0$ and $A = 0$. $\square$

Renormalisation remark. If a minimal-subtraction counterterm removes the first n Taylor terms, the per-mode integrand is the remainder $R_n(a) = \ln(1 + a) - \sum_{k=1}^n (-1)^{k-1} a^k / k$, with $R'_n(a) = (-1)^n a^n / (1 + a)$ single-signed and $R_n(0) = 0$, so $\text{sgn } R_n(a) = (-1)^n$ for $a \geq 0$ and $R_n(a) = 0$ iff $a = 0$. Hence any finite-order minimal-subtraction scheme keeps the renormalised trace a single-signed sum of per-mode terms vanishing only at $a = 0$, and the equality classification $\Delta F_{\text{ren}} = 0 \Rightarrow A = 0$ is scheme-independent. The only hypothesis beyond positivity is (H2) trace/determinant-class finiteness, exactly the finiteness already certified for ΔF .

7. Appendix B: sign-invariant covering partition

Lemma 3 (Sign-invariant covering partition). *Let $M(P)$ be the cell-averaged dressing variance of an admissible competitor $P \in \mathcal{A}_{\text{adm}}$. Suppose there exist levels $M_- \leq M_+$ such that: (i) for $M(P) \geq M_-$ the bulk bound holds, $F_{\text{layer}}[P] \geq F_{\text{layer}}[R_H] + m$ with $m > 0$ (Prop A) and $\delta F_{\text{off}}[P] > -m$ (STEP-5B), giving $F_{\text{total}}[P] > F_{\text{total}}[R_H]$; (ii) for $M(P) \leq M_+$ the near-gap floor holds, $\frac{1}{2} \text{Tr}[\ln(D_0 + \lambda' P^2) - \ln D_0] \geq 0$ (Appendix A), giving $F_{\text{total}}[P] \geq F_{\text{total}}[R_H]$. Then $\mathcal{A}_{\text{adm}} = \{M \geq M_-\} \cup \{M \leq M_+\}$ is a COVER and the selection inequality holds class-wide.*

Proof. When $M_- \leq M_+$ the two half-lines cover $\mathbb{R}$, so every P satisfies (i) or (ii); both yield $F_{\text{total}}[P] \geq F_{\text{total}}[R_H]$, strict off the sub-resolution quotient by the Appendix-A equality classification. $\square$

Sign-invariance. The verdict consumes only the overlap $M_- \leq M_+$: M_- is the smallest variance at which the Prop-A band margin $m > 0$ is certified, M_+ the largest at which the small-amplitude floor expansion is controlled; the certified Prop-A / STEP-5B data give $M_- \leq M_+$. The exact crossover $M_\times \in [M_-, M_+]$ is estimator-grade and enters neither bound's validity nor the sign; sharpening it to paper grade cannot change the verdict.

8. Result footer

Result ID:	B1-RH-ENUM (Reading-H class-wide selection)
Precise statement:	For every admissible competitor of the H-ADM-COH amended class, $F_{\text{total}}[P] \geq F_{\text{total}}[R_H]$ (strict on the sub-resolution quotient) at matched second-cumulant scope: Reading-H is selected. Proof = case split (bulk: layer margin Prop A + beyond-layer STEP-5B closure; near-gap: structural floor R-001/R-U10-3).
Scope:	Class-wide over the H-ADM-COH amended class; second-cumulant order; selection SIGN only. Quantitative margins estimator-grade (separate).
Hypotheses:	H-LAYER, H-ADM-COH, SC-SCOPE (all registered in claims/GATES.md).

Dependencies: A1-KERNEL-CONV (convention); B2-PROPA-HLAYER (Prop A, T6); B5 STEP-5B closure chain.

Evidence grade: ANALYTIC (assembled inequalities) + EXECUTED (STEP-5B 192/192; near-gap 14/14)

Reproduction command: python codes/vacuum/beyond_layer_gershgorin_bound.py && python codes/vacuum/neargap_common_mode_repair.py

Expected output: 192/192 and 14/14 PASS; the inequality chain $m > 0$, $dF_{\text{off}} > -m$, $\text{floor} \geq 0$ holds across the anchors.

Falsification gate: An admissible competitor with $F_{\text{total}} \leq F_{\text{total}}[R_H]$ (refutes selection); a near-gap section with the floor < 0 ; loss of any named hypothesis.

Tier before / after: T5 / T6 CONDITIONAL on {H-LAYER, H-ADM-COH, SC-SCOPE} (operator-authorized, "proceed with item 1" 2026-06-06).

No-overclaim statement: T6 is the selection SIGN modulo three named hypotheses; NOT unconditional, NOT all-orders (SC-SCOPE), NOT a precise-margin claim (ESTIMATOR-UPGRADE open), NOT μ^2 -robust (ROBUSTNESS-MU2 open). The unrestricted-class selection is the DR-2 route (research-grade).

Next required action: SC-SCOPE all-orders lift (third-cumulant endpoint kernels); ESTIMATOR-UPGRADE (controlled-error margins); ROBUSTNESS-MU2; then a T6 $\rightarrow$ higher reconsideration is a fresh operator decision.